Gradient Boosting Algorithm

1. Regression
2. Classification

Regression Dataset

Exp	Degree	Salary	$(\hat{y} - y)$ R_i
2	B.E	50K	−25K
3	Masters	70K	−5K
5	Masters	80K	5K
6	PhD	100K	25K

$\hat{y} = 75K$

Steps

1. Create a Base Model

→ compute the mean of the dependent feature. (in this case it will be 75)

2. Compute Residuals / Errors

→ subtract $\hat{y} - y_i$ to compute the residuals.

$$R_i = \hat{y} - y_i$$

Example

$$R_1 = 75K - 50K$$
$$= 25K$$

③ Decision Tree

→ consider inputs x_i and output as R_i to construct the decision tree.

$$F(x) = \alpha_0(h_0(x)) + \alpha_1(h_1(x)) + \alpha_2(h_2(x)) + \cdots + \alpha_n(h_n(x))$$

α – learning rate

$$\therefore \boxed{F(x) = \sum_{i=0}^{N} \alpha_i h_i(x)}$$

$\Downarrow$

Final function of Gradient Boosting